

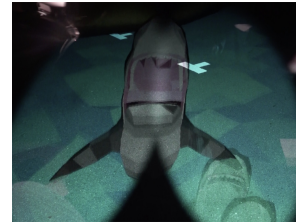
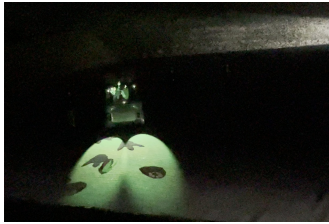
Development of 3D projection mapping from a moving vehicle to observe from inside and outside of the vehicle

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(a) A moving vehicle while projecting images on the ground in front of the vehicle



(b) The projected image observed from inside the vehicle
(c) The projected image observed from outside the vehicle

Figure 1: Overview of this study

KEYWORDS

3DCG, Projection Mapping, Moving Vehicle, Anamorphosis

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1 INTRODUCTION

In this study, we have developed a novel projection mapping system using a moving vehicle. In this system, two projectors are mounted on the vehicle and project images on the ground in front of the vehicle like headlights (Figure 1(a)). When the vehicle moves, the projected images change in synchronization with the movement of the vehicle. In addition, the images can be observed three-dimensionally from inside or outside of the vehicle (Figure 1(b)(c)). As a result, it is possible to represent a situation in which 3D objects existing in the dark appear illuminated by the headlights of a moving vehicle.

The principle of trick art and the principle of motion parallax are used for three-dimensional observation of the images. Therefore, it is not necessary to wear glasses when observing the images. With autonomous driving becoming more widespread, our projection mapping system could provide new entertainment for those traveling in the vehicle and looking at the vehicle.

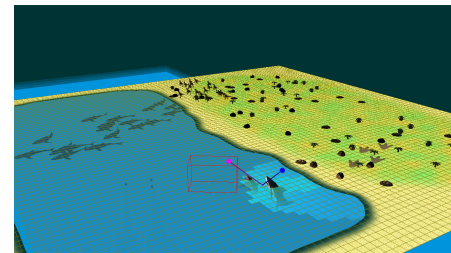


Figure 2: The 3DCG scene for generating projection images

2 METHODS

In the proposed projection mapping system, it is necessary to properly reflect the movement of the vehicle in projected images and to be able to observe the projected images three-dimensionally from the inside or outside of the vehicle. Therefore, the image for projection is generated based on 3DCG technology using the same method as our former study [Ahn and Mizuno 2021].

An area where the vehicle will travel is reproduced as a 3DCG scene (Figure 2). 2D or 3D objects which would be projected on the ground are placed in the 3DCG scene. The position and direction of the vehicle in the real world are acquired by GPS and compass, and they can be associated with the position and the direction in the 3DCG scene. The image to be projected on the ground is generated by setting the viewpoint and the viewing line in the 3DCG scene.

The proposed system uses the principle of trapezoidal anamorphosis, which is one of the trick arts, to realize three-dimensional observation of a 2D image. In anamorphosis, it is necessary to determine one viewpoint at which the observer can observe the 2D image three-dimensionally. In our projection mapping, the observer is assumed to be inside or outside of the vehicle.

When the observer is inside the vehicle, he/she stands inside and observes the image projected in front of the vehicle. Thus, the viewpoint and the viewing line of the observer move with the vehicle in the real world. The relative position and direction of

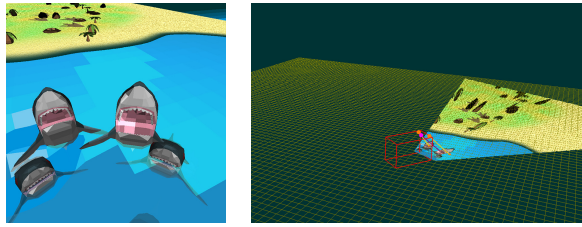
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(a) Rendering an image from the observer's viewpoint (inside the vehicle) (b) Projecting and mapping the image (a) onto the ground in the 3DCG scene

Figure 3: Reproducing a projected image for three-dimensional observation from the observer's viewpoint

the observer to the vehicle are fixed inside the vehicle, and the absolute position and direction of the viewpoint and the viewing line of the observer are calculated by combining with the vehicle's position information. When the observer is outside the vehicle, the viewpoint of the observer is fixed, while the viewing line moves with the vehicle to face the projected image in the real world.

The viewpoint and the viewing line of the observer in the real world can be mapped to those in the 3DCG scene. The system renders a CG image of 2D/3D objects placed in the 3DCG scene based on the viewpoint and the viewing line determined in the 3DCG scene (Figure 3(a)). Then the rendered image is projection-mapped onto a plane in the 3DCG scene corresponding to the ground in the real world (Figure 3(b)). The mapping result shows a projected image that is observed three-dimensionally by the principle of anamorphosis when viewed from the observer.

The system reproduces the mapping results of the 3DCG scene by projecting images onto the ground from projectors mounted on the vehicle. To generate the projection images, 2D/3D objects are removed and only the image-mapped ground is left in the 3DCG scene. Since, the relative positions and the directions of two projectors mounted on the vehicle can be measured, their absolute positions and directions are calculated and mapped to the viewpoints and viewing lines in the 3DCG scene, just as in the case of the observer's viewpoint and viewing line. Then, using each viewpoint and each viewing line with the same view volumes as the projectors, two CG images of the 3DCG scene with the image-mapped ground are rendered. These images are projected on the ground from each projector mounted on the vehicle.

3 EXPERIMENT

We conducted an experiment using a vehicle that can acquire position information for automated driving. The system was implemented on a MacBookPro using C++ and OpenGL. Two mobile projectors (700(ANSI lumen)) were placed at a height of 1.8(m) from the ground in front of the vehicle and angled to illuminate the ground like headlights. The 3DCG scene was inspired by the age of dinosaurs, with small/hand-drawn [Ahn and Mizuno 2021] dinosaurs running and the legs of tyrannosaurs appearing on the land, and fish swimming and sharks showing their faces in the ocean. The point inside the vehicle to observe the images three-dimensionally by the anamorphosis effect was set at the center of the vehicle at a height of 1.6(m) from the floor, and the point outside the vehicle was set at a height of 1.6(m) from the ground, 2(m) away

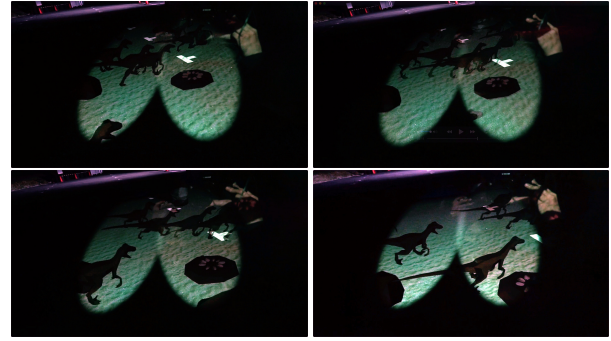


Figure 4: Observing dinosaurs from inside of the vehicle



Figure 5: Observing sharks from inside of the vehicle



Figure 6: Observing sharks from outside of the vehicle

from the vehicle's running route. The vehicle moved at a maximum speed of 5(km/h) in an area of approximately 40(m) square, and the position information was applied to the 3DCG scene at 30(fps).

Figure 4 and Figure 5 show the results of observing the projected image from inside the vehicle. The image changed properly in synchronization with the movement of the vehicle, and dinosaurs and sharks in the darkness appeared illuminated by the headlights and were observed three-dimensionally. Figure 6 shows the results of observing the projected images from outside the vehicle. By standing at a pre-defined position, sharks were observed three-dimensionally as they emerged from the water when the vehicle moved and the darkness was illuminated.

Creating projection contents which people both inside and outside the vehicle can enjoy simultaneously is one of feature works.

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Sora Ahn and Shinji Mizuno. 2021. "Sketch Dance Stage Online": Three-dimensional CG projection of hand-drawn characters to real space and interaction. In *SIGGRAPH ASIA 2021 Posters*.